

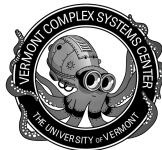
Spatial epidemiology and adaptive targeted sampling to manage domestic Triatomine infestations in Guatemala

B. K. M. Case

2 August 2022

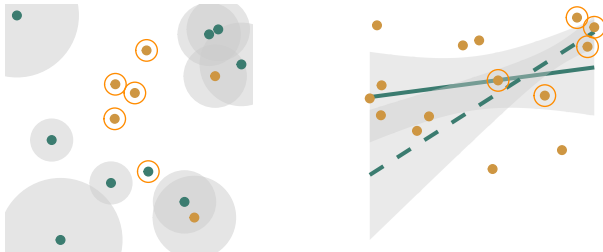


The University of Vermont



Targeted sampling initiatives for public health

- Tradeoff between **sampling to mitigate risk** and **sampling for inference**
- Leads to *selection bias*, which makes diagnosing risk less accurate
- A problem of *exploration vs. exploitation* in **design space**¹
 - **Targeted control**: more exploitation than random
 - **Targeted surveillance**: more exploration than random



¹Nichols et al., “Strategic testing approaches for targeted disease monitoring can be used to inform pandemic decision-making”, *PLOS Biology*. 2021.

Control for Triatomine infestations

- Keeping community infestation low can eliminate *T. cruzi* seroprevalence²
 - Infestation prevalence below 5% a common target³
- Complicated by limited resources/personnel, and by variability in vector ecology, socioeconomic risk factors, and cultural practices^{4,5}
 - Efficient and localized strategies are required
- Targeted sampling could help two key areas:
 - 1 **Control.** Identify infested houses for treatment
 - 2 **Surveillance.** Assess 5% threshold throughout the community

²Aiga et al., "Chagas disease: assessing the existence of a threshold for bug infestation rate", *American Journal of Tropical Medicine and Hygiene*. 2012.

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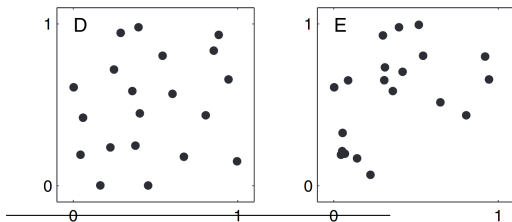
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Geostatistical design

- Given spatial phenomena Z over domain $\mathcal{D}$, choose discrete set $\mathcal{S} \subset \mathcal{D}$ to minimize variance over $\mathcal{D} \setminus \mathcal{S}$
 - Maximum or average variance can be used
- Tradeoff between prediction and inference of spatial parameters⁶
 - Naturally accounted for in Bayesian paradigm⁷
 - Particularly attractive for non-linear models and small $\mathcal{S}$
- Finding an optimal $\mathcal{S}$ is infeasible even for moderate-size $\mathcal{S}$
 - Heuristics such as "lattice plus close pairs" are popular alternative



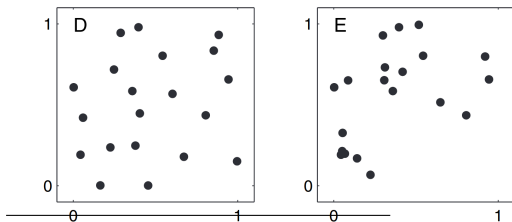
Classical space-filling design (left) vs. Bayesian approach (right), for minimizing spatially-averaged variance. From Diggle and Lophaven (2006)

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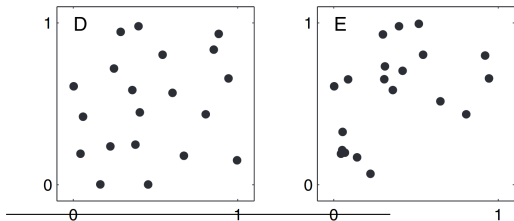
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Adaptive geostatistical sampling

- Initially proposed for spatial statistics by Chipeta et al. (2016)⁸ for Malaria surveys,⁹ where spatially-averaged variance was design target
- “Sharpens” the space of high-quality designs
- Flexibility** makes it well-suited for surveys in public health

Adaptive sampling algorithm

- 1 Sample initial design $\mathcal{S}_1$ uniformly and set $i = 1$
- 2 Fit the posterior distribution $p(\theta | \mathbf{y}_i)$
- 3 Assign each location $\mathbf{s}_0 \in \mathcal{D} \setminus \mathcal{S}_i$ a utility $U(\mathbf{s}_0)$
- 4 Choose the b locations with highest utility and add them to $\mathcal{S}_i$
- 5 Set $i = i + 1$ and repeat steps 2-5 until done.

⁸Chipeta et al., “Adaptive geostatistical design and analysis for prevalence surveys”, *Spatial Statistics*. 2016.

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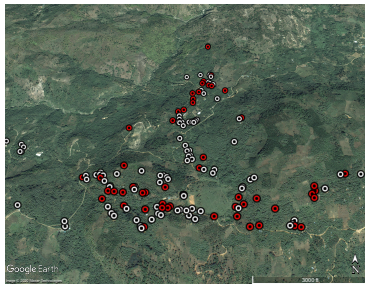
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Our contribution

- Can adaptive geostatistical sampling be used to identify a **minimal number of houses** for treatment while still **meeting the reduction target** throughout the community?
- Develop a class of utility functions that transitions from **exploring the entire design space** to **targeting areas of high risk**

Overview

- Data were collected from five villages in Chiquimula, Guatemala from August-December 2011^{10,11}
- Socioeconomic variables included number/type of domestic animals, housing materials, clutter around the home



¹⁰Bustamante et al., "Information to act: household characteristics are predictors of domestic infestation with the Chagas vector *Triatoma dimidiata* in Central America", *American Journal of Tropical Medicine and Hygiene*. 2015.

¹¹Lima-Cordón et al., "Implementation science: epidemiology and feeding profiles of the Chagas vector *Triatoma dimidiata* prior to EcoHealth intervention for three locations in Central America", *PLoS Neglected Tropical Diseases*. 2018. [◀](#) [≡](#) [▶](#) [≡](#) [↺](#) [↻](#)

Observation model/data likelihood

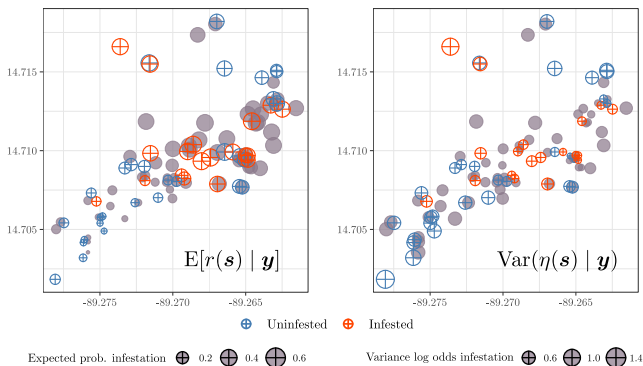
For each house $\mathbf{s} \in \mathcal{D}$, assume infestation status $y(\mathbf{s})$ follows a logistic regression model with spatial and non-spatial random effects:

$$y(\mathbf{s}) \sim \text{Bernoulli}(r(\mathbf{s}))$$
$$\text{logit}(r(\mathbf{s})) = \eta(\mathbf{s}) = \mathbf{x}(\mathbf{s})\boldsymbol{\beta} + z(\mathbf{s}) + \varepsilon(\mathbf{s}),$$

- r : risk of infestation
- η : linear predictor
- $\mathbf{x}$: measured covariates
- $\boldsymbol{\beta}$: fixed effect coefficients
- z : residual spatial variation
- ε : residual non-spatial variation

Parameter inference using INLA

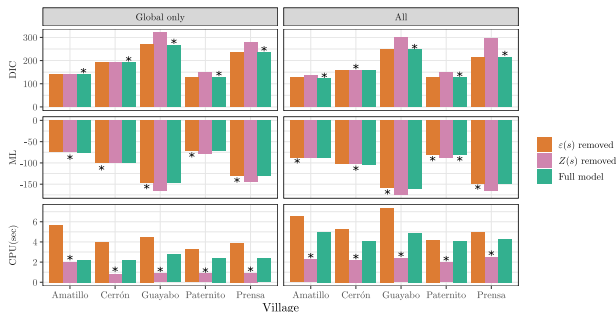
- Fit posterior marginals using Integrated Nested Laplace Approximation¹²



¹²Lindgren and Rue, "Bayesian spatial modelling with R-INLA", *Journal of Statistical Software*, 2015.

Full-village analysis

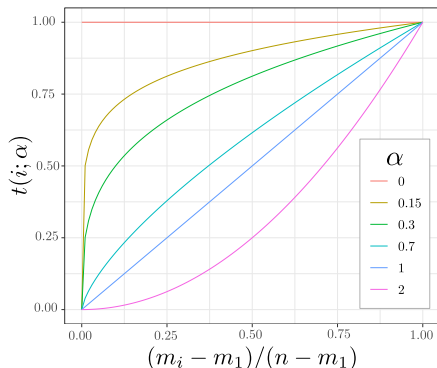
- Compared to models without spatial/non-spatial effects
- Compared full covariate set (“all”) and those available with GIS (“global”)
 - Lots of variety in fixed effect coefficients



Choosing new locations

$$U(\mathbf{s}_0) = t(i; \alpha) \tilde{r}(\mathbf{s}_0) + (1 - t(i; \alpha)) \tilde{\nu}(\mathbf{s}_0)$$

- **Sample infested locations**
→ use normalized risk $\tilde{r}(\mathbf{s}_0)$
- **Minimize sampling bias**
→ use normalized variance $\tilde{\nu}(\mathbf{s}_0)$
- Interpolate with *transition function* $t(i; \alpha)$



Termination condition

- Interested in state of whole village, not just individual homes
- Need the joint distribution of unvisited houses, $\Pr(\hat{\mathbf{y}}_0 | \mathbf{y})$
 - Available from R-INLA, may get faster in the future¹³

	$\hat{y}_1$	$\hat{y}_2$	$\hat{y}_3$	$\dots$
$\hat{\mathbf{y}}_0^{(1)}$	1	0	0	$\dots$
$\hat{\mathbf{y}}_0^{(2)}$	1	1	1	$\dots$
$\hat{\mathbf{y}}_0^{(1)}$	0	1	1	$\dots$
$\vdots$		$\vdots$		$\ddots$

- Use to approximate the *termination condition*

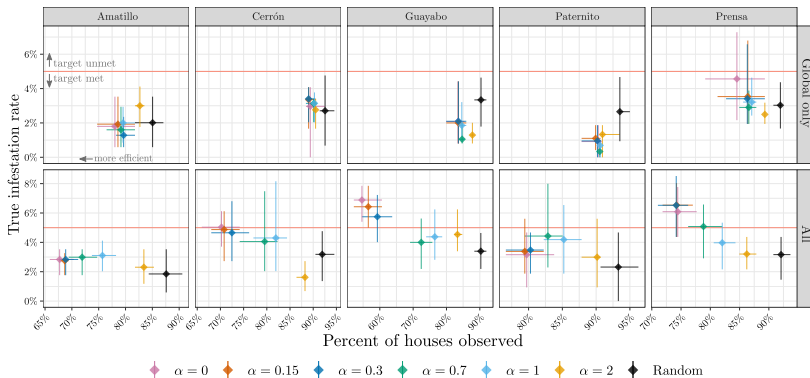
$$\Pr(\mathbf{1} \cdot \hat{\mathbf{y}}_0 < \kappa n) \geq \gamma,$$

for *reduction target* κ and *confidence level* γ

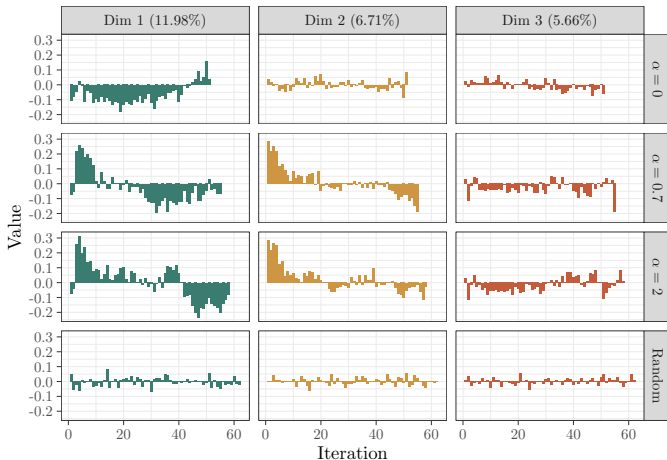
¹³Chiuchiolio, Niekerk, and Rue, "Joint Posterior Inference for Latent Gaussian Models with R-INLA", *arXiv preprint arXiv:2112.02861*. 2021

Simulation study

- For each village, select 50 random initial designs of 10 houses



Exploration behavior



Mean coordinate values across replications in La Prensa, using Multiple Correspondence Analysis.

Practical considerations

- Socioeconomic variables impractical for initial control campaign, but not for ongoing surveillance
- Sampling constraints
 - Centralizing information between iterations
 - Distance between locations and individualized routes
 - Collaboration with individuals collecting data¹⁴
- Preferable settings for α depend on the intervention
 - Unclear why optimal α varies between villages and covariate sets
 - May change throughout a large-scale intervention

¹⁴Gutfraind et al., "Integrating evidence, models and maps to enhance Chagas disease vector surveillance", *PLOS Neglected Tropical Diseases*. 2018.

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Correcting preferential sampling estimates

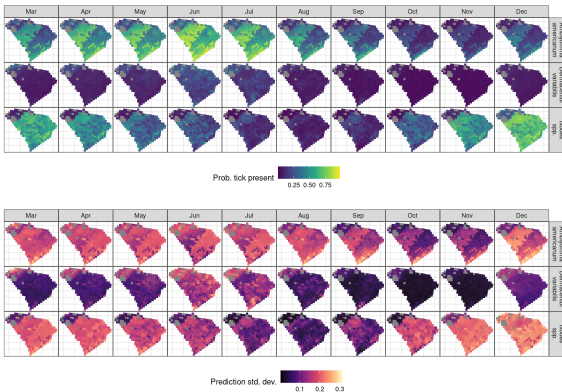
- Possible to correct population-level statistics under preferential sampling^{15,16}
 - Requires statistical model for sampling locations
 - Could improve the accuracy of assessing whether control target is met

¹⁵J. Zidek, Shaddick, and Taylor, "Reducing estimation bias in adaptively changing monitoring networks with preferential site selection", *Annals of Applied Statistics*. 2014.

¹⁶Watson, J. V. Zidek, and Shaddick, "A general theory for preferential sampling in environmental networks", *The Annals of Applied Statistics*. 2019.

Prospective sampling

- Select new locations based on *anticipated* variance reduction after $\mathbf{y}_{\text{new}}$ would be observed
- Involves additional integral over $p(\hat{\mathbf{y}}_{\text{new}} \mid \mathbf{y})$



Acknowledgements

Thank you to my collaborators, and to the researchers, Ministry of Health personnel, and people of Chiquimula who participated in data collection.

Paper:

Case BKM, Young JG, Penados D, Monroy C, Hébert-Dufresne L, Stevens L. "Spatial epidemiology and adaptive targeted sampling to manage the Chagas disease vector *Triatoma dimidiata*", *PLOS Neglected Tropical Diseases*. 16(6): e0010436.

<https://doi.org/10.1371/journal.pntd.0010436>.

Code:

<https://github.com/brendandaisy/chagas-adaptive-sampling/tree/plos-ntd>